

Steering System:

- during taxiing is achieved by differential braking or tail rotor thrust

Tail wheel lock:

Tail landing gear is fitted with a pilot operated tail wheel lock through a control cable

Day 10Limitations

- ambient temp range : -40°C to $+55^{\circ}\text{C}$
- pilot in front, weapon system operator at rear (wso)
- Take off and landing on slope greater than 3°
- Spot turns (turns whilst hovering) with rate greater than $60^{\circ}/\text{sec}$
- Sideward flight / hover in cross winds beyond 50 km/h
- Rearward flight or hover in tailwinds beyond 50 km/h

Critical wind azimuth?(ix) slope landing limitations

- permitted upto 3° slope in all directions
- Avoid exceeding mast moment limit of 37 KN-m during landing on slopes
- Sloped up to the left and forward of the helicopter entails least degree of difficulty

Vortex ring Conditions

- Vertical descent very close to 0 - free from vortex ring condition up to descent rate of 150 m/min
- at descent rates of 200 m/min & beyond, - increase in vibration, random increase in pitch, roll & yaw requiring increased control activity indicate incipient vortex ring conditions

Turret
Gun
Rocket
air to air missile

classmate

Date

Page

Rotor starting & stopping

Starting & stopping executed in surface wind conditions

45 Kts (85 Km/h) Head wind

27 Kts (50 Km/h) Side wind

17 Kts (30 Km/h) tail wind

weights & loading limits

- The helicopter is cleared for op up to max of 80% of gross weight

Landing light limitation:

maximum duty cycle is 15 min on, followed by 15 min off

Range marking symbology:

Red - warning range

Yellow - caution range

Green - normal operating range

△ > reds - min / max Transient limit / SER limit

Red line / dashed line - min / max limit or MCR limit

Yellow line - min / max normal operating limits

Emergency landing:

1) land immediately:

land in an area that will have least negative impact on crew survivability

2) land ASAP:

land at nearest adequate landing site at which a safe approach & landing can be made

3) Divert: divert to the nearest suitable helipad or air field

- 4) Return to Base : return to operating base, continue to destination, any other air field / helipad, which has support maintenance facilities.

Autorotative landing

Entry to autorotation

Collective : fully down

Forward speed : adjust to 120-150 km/h

Rotor RPM : adjust b/w 95-105%

Autorotative landing:

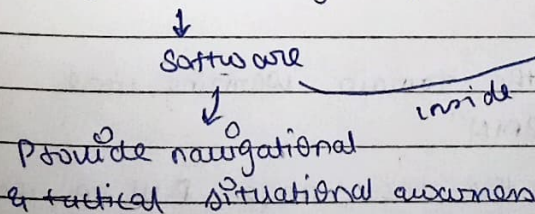
Rotor RPM : maintain 95-105%

at 40-30 m agl : gentle flare to about 15° - 20° nose up raising collective to keep NR within limits

at 7-5 m agl : Reduce frame attitude slightly to about 10° - 15° nose up touch down attitude

Landing on slopes - mast moments high, - after touchdown AFCS is disengaged & mast moments adjusted within limits

The Digital moving map (DMM) and display & mission computer (DMC)



2D FLV, 2D PLAN, 3D LOS, 3D SIM, 3D GFW

2D FLV :

aircraft position - aircraft path over predefined selected terrain

2D PLAN :

- define new routes, waypoints, Zaimark certain areas by use of various shapes

3D LOS:

- used to provide line of sight (LOS) data around a point in a predefined radius
- LOS data is displayed based on the elevation of the selected point on ground, the present altitude of the aircraft & in the selected radius

3D SIM:

simulate aircraft flight over predefined course / any predetermined point on a particular bearing can be simulated & observe from 3D perspective

Default digital terrain elevation data (DTED) loaded in DNM.

3 backgrounds: MAP, SAT, BLK

MAP - navigational background

SAT - Satellite photo as navigational background

BLK - black background

- ~~Map~~ GPWS-on:

Obstacle within the terrain warning time
"TERRAIN"

Obstacle within the LOS range, Pull up warning is displayed along with voice warning

- No communication b/w DNM & DMC, map freezes

Light Utility Helicopter (LUH)

- 3 ton class, single turbo shaft engine
- skid type landing gear - forward cross tube, rear cross tube & 2 skid tubes
- 4 blades

Seating : 2 pilots + 6 passengers
 max take off weight : 3000 kg

Transmission :

main gearbox :

input speed (Engine - N1/B 5000) - 6000 RPM

main motor drive speed - 307 RPM

Tail gear box :

1/r speed : 4003 RPM

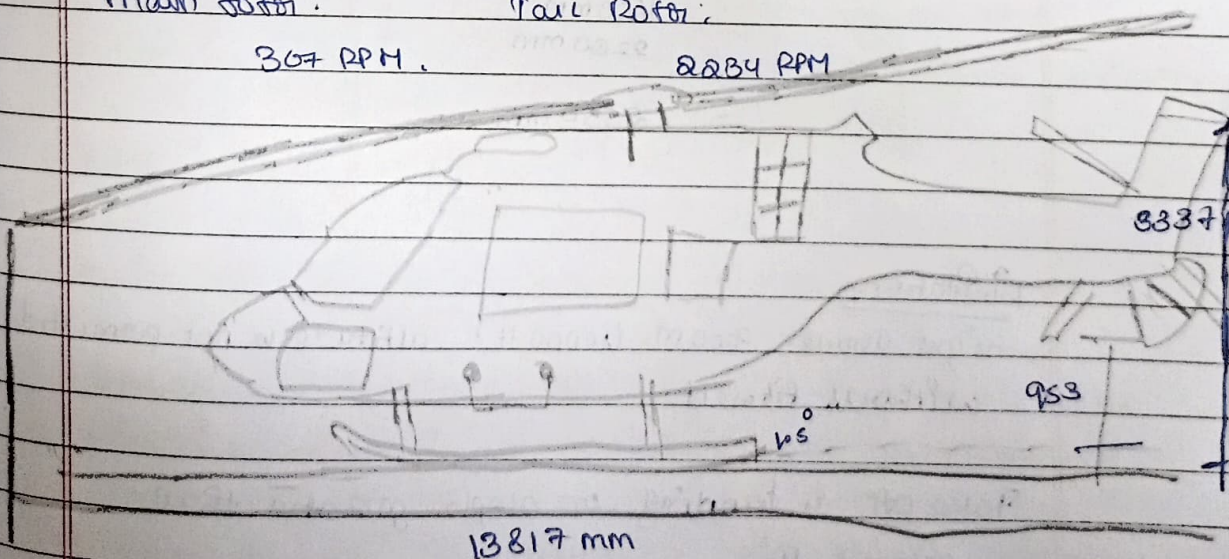
Tail motor drive speed : 2234 RPM

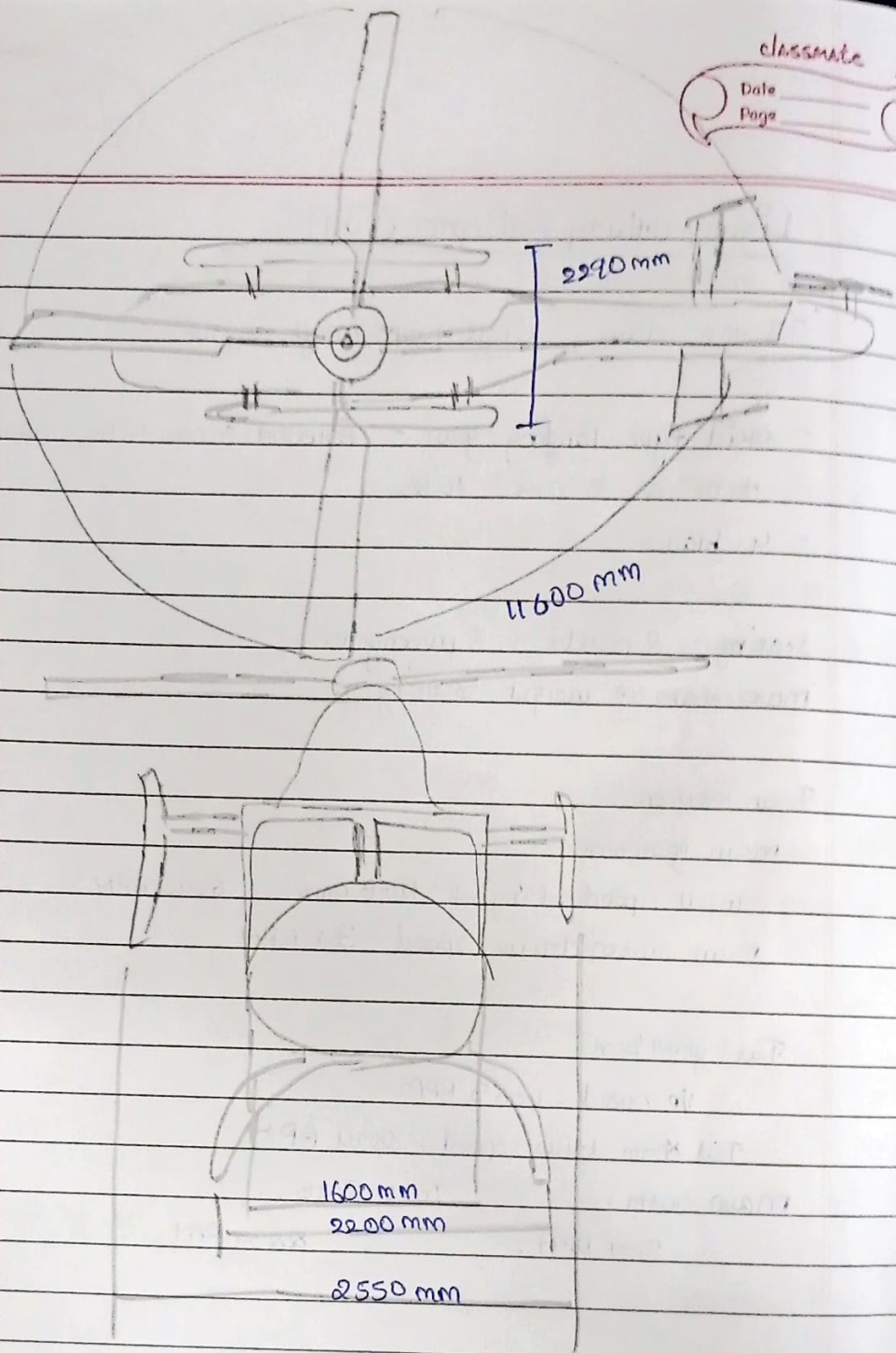
Main rotor :

307 RPM

Tail Rotor :

2234 RPM





Limitations :-

flight above 3000 m (10000 ft) altitude is not permitted without permit

Take off & landing on slopes greater than

upslope - 10°

down slope = 5°

lateral slope = 6°

spot turns with greater than $60^\circ / \text{sec}$

sideward flight or hover in cross wind beyond 15 kt

Reverse flight & hover in tailwinds beyond 17kt

slope landing limitations:

Take off and landing with a nose down attitude should be performed with caution and only when terrain permits adequate clearance before blow back & ground

ground sloping down to the left - 6°

ground sloping down to the right - 6°

ground sloping nose up - 10°

ground sloping nose down - if tail clearance allows - 5°

max achievable slope angle may be further limited by character & friction of the landing surface

when the altitude indicator is used to determine the slope, the normal attitude on ground of 1.5° up shall be considered

5° nose up slope - 6.5° nose up on attitude indicator

5° nose down slope - 3.5° nose down on attitude indicator

Vertical Sling conditions

Avoid exceeding 400ft/min rate of descent at MS close to 0

Vertical descent at speeds very close to 0 has been found to be free from the vortex sling condition up to the rate of descent of 400ft/min

descent rate 600ft/min and beyond - increase in vibration along with random pitch, roll & yaw requiring increased control activity indicate incipient vortex sling conditions

Rotor start & stop in strong winds

35 knots (65 km/h) Head wind

35 knots (65 km/h) Side wind

35 knots (65 km/h) tail wind

landing gear limitation:

max take off/landing weight = 3150 kg

max vertical velocity = 2.54 m/s

max forward velocity for landing = 40 kt